Ai Ml Notes

Artificial Intelligence (AI)

Artificial Intelligence is a field of computer science that focuses on building systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, language understanding and decision making. AI can be classified into narrow AI and general AI.

Machine Learning (ML)

Machine Learning is a subset of Artificial Intelligence in which computers learn patterns from data instead of being explicitly programmed for every task. Common types include supervised learning, unsupervised learning and reinforcement learning. ML is used for prediction, classification and recommendation.

Supervised Learning

Supervised learning uses labelled training data. The model learns the relationship between input data and known output labels. Examples include spam detection, house price prediction and image classification.

Unsupervised Learning

Unsupervised learning works with unlabelled data and discovers hidden patterns or groups. Clustering and dimensionality reduction are common unsupervised learning tasks.

Deep Learning

Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers. It is especially useful for complex data such as images, audio and text. Popular applications include computer vision, speech recognition and natural language processing.

Neural Networks

A neural network is a computational model inspired by the human brain. It consists of an input layer, one or more hidden layers and an output layer. Neurons process inputs using weights and activation functions, and the model learns by adjusting its weights during training.

CNN - Convolutional Neural Network

A Convolutional Neural Network is a deep learning architecture mainly used for image and computer vision tasks. CNNs use convolution layers to extract visual features such as edges, textures and shapes. Pooling layers reduce spatial dimensions, while fully connected layers help produce final predictions.

RNN - Recurrent Neural Network

A Recurrent Neural Network is designed for sequential data. It uses information from previous steps when processing new input. RNNs have been used for text, speech and time-series data. Variants such as LSTM and GRU help handle long-term dependencies.

Overfitting

Overfitting happens when a model learns the training data too closely, including noise, and performs poorly on new data. Techniques such as regularization, dropout, data augmentation and using more training data can help reduce overfitting.

Underfitting

Underfitting occurs when a model is too simple to learn the important patterns in the training data. Both training and testing performance may be poor. Increasing model capacity or improving features can help.

TensorFlow and PyTorch

TensorFlow and PyTorch are popular deep learning frameworks. They provide tools for building, training and deploying neural network models.